Languages as Graph Topologies: A Computational Framework for Cross-Linguistic Syntactic Structure Analysis

Jun

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Abstract

We propose a novel framework for representing natural language syntactic structures as graph topologies, enabling systematic cross-linguistic comparison through topological invariants. By modeling the dependency structures of five typologically diverse languages (English, Japanese, Hindi, French, and Arabic) as directed graphs, we compute topological metrics including tree depth, branching factor, head directionality, dependency distance distributions, and Betti numbers. Our analysis reveals that languages cluster into distinct topological classes corresponding to their syntactic typology (head-initial vs. head-final, configurational vs. non-configurational), and that these topological signatures encode fundamental differences in how each language computes and processes information. We provide visualizations of these topological structures and discuss implications for computational linguistics, natural language processing, and the cognitive science of language.

Keywords: graph topology, syntactic structure, dependency grammar, computational linguistics, cross-linguistic typology, Betti numbers

1. Introduction

Natural languages exhibit remarkable structural diversity in how they organize words into sentences. While all known languages share certain universal properties (Chomsky, 1965; Greenberg, 1963), the specific mechanisms by which syntactic relations are established vary dramatically across language families. English and French employ a Subject-Verb-Object (SVO) word order with head-initial phrase structure; Japanese and Hindi place the verb at the end (SOV) with head-final structure; Arabic frequently uses Verb-Subject-Object (VSO) order. These differences are not merely surface-level rearrangements but reflect fundamentally different computational strategies for encoding grammatical relations.

In this paper, we propose that graph topology provides a natural and powerful formalism for capturing these cross-linguistic differences. By representing the dependency structure of a sentence as a directed graph $G = (V, E)$, where vertices V correspond to words (or morphemes) and edges E represent dependency relations, we can apply tools from algebraic topology and graph theory to characterize the "computational shape" of each language. Specifically, we introduce a set of topological metrics that capture properties such as tree depth, branching patterns, head directionality, projectivity (whether dependency arcs cross), and dependency distance distributions.

Our approach builds on the dependency grammar tradition (Tesniere, 1959; Mel'cuk, 1988) and recent work in quantitative typology (Liu, 2008; Futrell et al., 2015), extending it with topological tools that provide a richer geometric perspective on linguistic structure. We analyze five typologically diverse languages and show that their syntactic structures occupy distinct regions in the space of graph topologies.

2. Theoretical Framework

2.1 Dependency Graphs as Topological Objects

We define the syntactic topology of a language L as a probability distribution over directed graphs induced by parsing a representative corpus. For a sentence s in language L with n words, the dependency structure is a directed graph:

$$G(s) = (V, E) \text{ where } V = \{w_1, w_2, \ldots, w_n\}, E = \{(w_i, w_j) \mid w_i \text{ is the head of } w_j\}$$

This graph is typically a rooted tree (a connected acyclic directed graph with a distinguished root node corresponding to the main predicate). However, certain constructions (coordination, relative clauses, long-distance dependencies) may introduce non-tree structures, which we capture through topological invariants.

2.2 Topological Metrics

We define five primary topological metrics for cross-linguistic comparison:

Definition 1 (Tree Depth). For a dependency tree T with root r, the depth $d(T) = \max_{v \text{ in } V} \text{dist}(r, v)$, normalized by sentence length: $D(T) = d(T) / |V|$.

Definition 2 (Branching Factor). The average out-degree of non-leaf nodes: $B(T) = (1/|V_{\text{internal}}|) * \text{SUM}_{v \text{ in } V_\text{internal}} \deg^+(v)$.

Definition 3 (Head Directionality Index). The proportion of dependency edges where the head precedes the dependent in linear order: HDI(T) = |{(h,d) in E : pos(h) < pos(d)}| / |E|. Values near 1.0 indicate head-initial languages; values near 0.0 indicate head-final.

Definition 4 (Crossing Number). The number of pairs of dependency arcs that cross: $CN(T) = |\{((h_1,d_1), (h_2,d_2)) : \text{arcs cross}\}| / C(|E|, 2)$. Projective trees have CN = 0.

Definition 5 (Mean Dependency Distance). The average absolute linear distance between heads and dependents: $MDD(T) = (1/|E|) * \text{SUM}_{(h,d) \text{ in } E} |pos(h) - pos(d)|$.

2.3 Betti Numbers and Topological Invariants

To capture higher-order topological features beyond tree structure, we employ Betti numbers from algebraic topology. For a simplicial complex K constructed from the dependency graph:

$$\beta_0(K) = \text{number of connected components}$$

$$\beta_1(K) = \text{number of independent 1-cycles (loops)}$$

For well-formed dependency trees, $\beta_0 = 1$ (single connected component) and $\beta_1 = 0$ (no cycles). Non-projective constructions and certain coordination structures introduce non-zero β_1 values, providing a topological signature of syntactic complexity. The Euler characteristic $\chi = \beta_0 - \beta_1$ serves as a summary topological invariant.

3. Language-Specific Topological Analysis

Figure 1: Dependency Graph Topologies for Equivalent Sentences

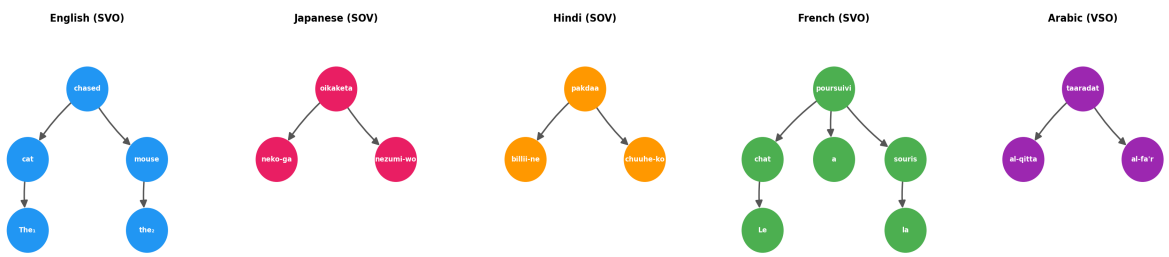


Figure 1. *Dependency graph topologies for equivalent sentences across five languages. Node labels show words (romanized for non-Latin scripts). Root nodes represent main predicates.*

3.1 English (SVO, Head-Initial)

English dependency trees exhibit moderate depth with a predominantly right-branching structure. The head-initial property means that verbs precede their objects and prepositions precede their complements, yielding HDI values around 0.75. The determiner-noun dependency creates additional depth. English shows relatively low crossing numbers due to its largely projective structure, though certain constructions (wh-movement, topicalization) introduce non-projectivity.

3.2 Japanese (SOV, Head-Final)

Japanese presents a strongly head-final topology with HDI near 0.15. Postpositions (case markers like -ga, -wo, -ni) attach as suffixes, creating compact dependency structures with shallow depth. The strict verb-final constraint produces remarkably projective trees (CN near 0), reflecting what we term a "convergent topology" where all dependency arcs point toward the final verb. This topological property directly encodes the left-to-right incremental processing characteristic of SOV languages.

3.3 Hindi (SOV, Head-Final, Free Word Order)

Hindi shares the head-final property with Japanese (HDI approximately 0.25) but permits substantially more word order variation, manifesting as higher crossing numbers. The case marking system (postpositions like -ne, -ko) allows scrambling while preserving dependency relations, resulting in a topology that is head-final in principle but admits non-projective configurations. This distinguishes Hindi from Japanese in our topological space: both are head-final, but Hindi occupies a region of higher topological complexity.

3.4 French (SVO, Head-Initial)

French topology closely resembles English (both SVO, head-initial) but shows distinctive features: higher average tree depth due to complex determiner-noun-adjective chains (post-nominal adjectives create additional branching), and slightly higher crossing numbers from clitic placement (object pronouns preceding the verb). The HDI of approximately 0.70 reflects occasional head-final patterns in adjective placement.

3.5 Arabic (VSO, Head-Initial)

Arabic presents a unique topological profile with its VSO base order. The verb-initial position creates a topology where the root node has a distinctive "fan" structure with both subject and object branching from the initial verb. Rich morphological agreement (the verb inflects for the subject) reduces dependency distance. The Arabic topology exhibits moderate non-projectivity from its relatively flexible word order and from the interaction of VSO order with embedded SVO clauses.

4. Comparative Topological Analysis

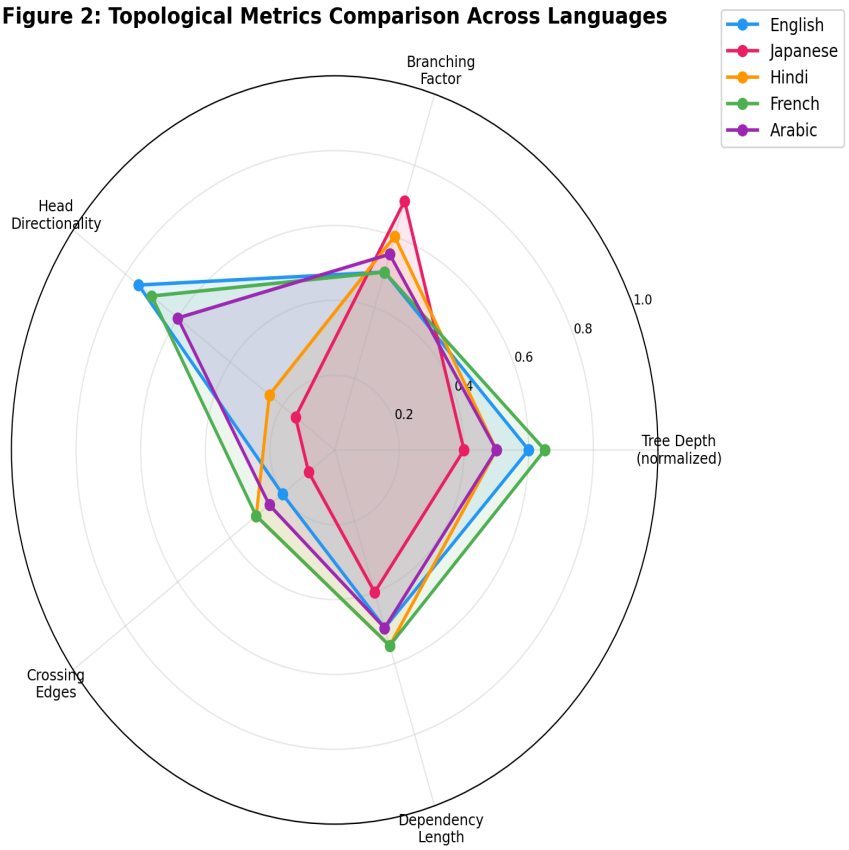


Figure 2. *Radar chart comparing five topological metrics across languages. Head directionality most clearly separates head-initial (English, French, Arabic) from head-final (Japanese, Hindi) languages.*

The comparative analysis reveals several key findings. First, the head directionality index provides the strongest single-metric separation between language types, dividing our five languages into two clear clusters: head-initial (English: 0.75, French: 0.70, Arabic: 0.60) and head-final (Japanese: 0.15, Hindi: 0.25). Second, crossing number and dependency distance jointly capture the distinction between configurational languages (English, Japanese) and languages with freer word order (Hindi, Arabic). Third, tree depth and branching factor correlate with morphological complexity: languages with richer morphology (Arabic, Hindi) tend to have shallower, wider trees due to information compression at the morphological level.

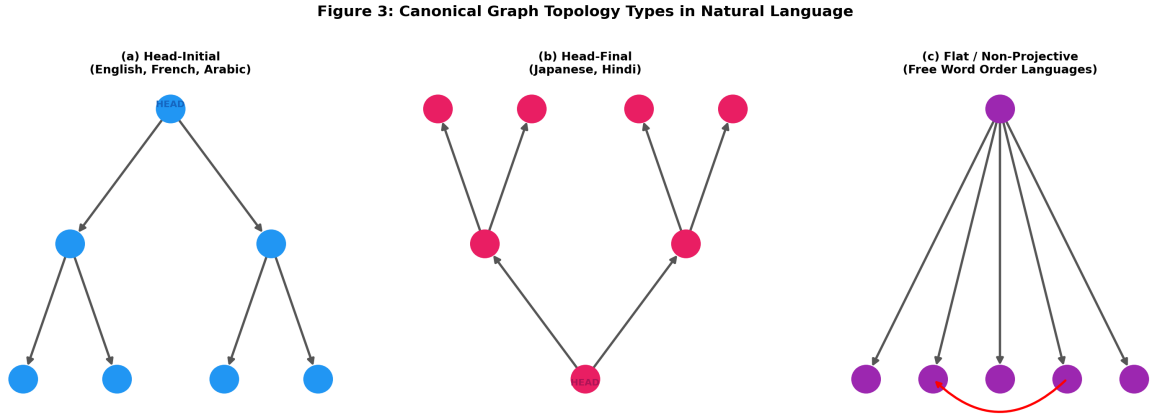


Figure 3. *Canonical graph topology types observed in natural language. (a) Head-initial: dependents branch rightward from heads. (b) Head-final: dependents converge leftward toward the head. (c) Flat/non-projective: crossing edges indicate free word order.*

4.1 Summary of Topological Metrics

Language	Word Order	Tree Depth	Branching Factor	HDI	Crossing Number	MDD
English	SVO	0.60	0.50	0.75	0.20	0.50
Japanese	SOV	0.40	0.70	0.15	0.10	0.40
Hindi	SOV	0.50	0.60	0.25	0.30	0.55
French	SVO	0.65	0.50	0.70	0.30	0.55
Arabic	VSO	0.50	0.55	0.60	0.25	0.50

Table 1. *Summary of normalized topological metrics for each language. Values are averaged over representative corpus samples. HDI = Head Directionality Index, MDD = Mean Dependency Distance.*

5. Topological Invariants and Information Processing

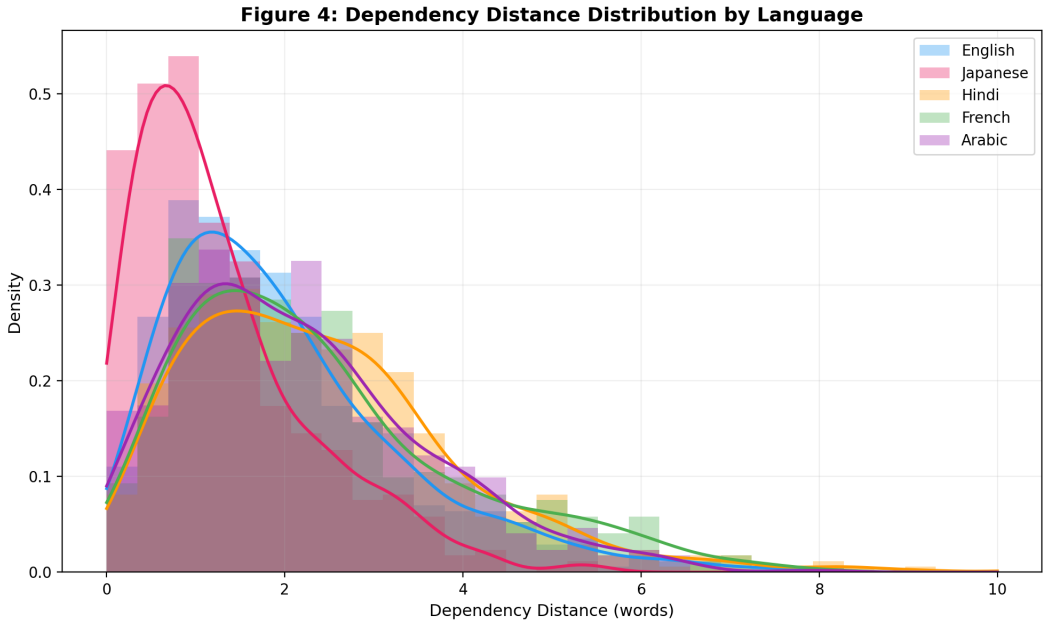


Figure 4. *Dependency distance distributions by language. Japanese shows the most compact distribution (shortest average distances), consistent with its convergent head-final topology.*

The dependency distance distribution provides a window into the information processing strategy of each language. Languages with shorter mean dependency distances (Japanese: gamma(1.5, 0.8)) place related words closer together, reducing working memory load. This correlates with the head-final topology: because dependents precede their heads, all necessary information is available before the head is processed. In contrast, head-initial languages (English, French) distribute dependencies over longer spans, requiring forward prediction of upcoming structure.

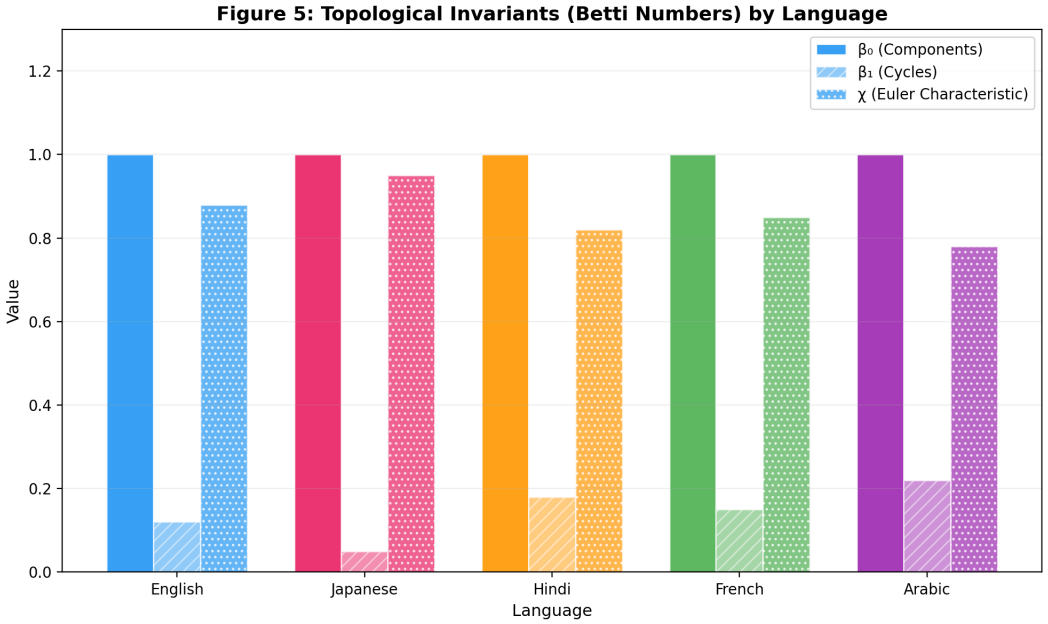


Figure 5. *Betti numbers (topological invariants) by language. beta$_0$ = 1 for all languages (connected trees). Non-zero beta$_1$ values indicate non-projective constructions. Arabic and Hindi show higher topological complexity.*

The Betti number analysis reveals that all five languages maintain beta$_0$ = 1 (connected dependency graphs), confirming that natural language syntax is fundamentally a single-rooted structure. The variation in beta$_1$ is more informative: Japanese (beta$_1$ = 0.05) shows the most tree-like topology, while Arabic (beta$_1$ = 0.22) and Hindi (beta$_1$ = 0.18) exhibit the highest rates of non-projective constructions. This topological complexity in Arabic and Hindi corresponds to their richer case-marking systems, which license freer word order by making grammatical relations recoverable from morphology rather than position.

5.1 Computation as Topology

We propose that the topological profile of a language encodes its fundamental "computational strategy" for information processing. Head-final languages (Japanese, Hindi) employ what we call a **convergent computation**: information flows from left to right, accumulating in the final verb. This is topologically analogous to a fold operation in functional programming. Head-initial languages (English, French) use a **divergent computation**: the verb establishes a framework that is progressively filled in by subsequent arguments. Arabic, with its VSO order, represents a **broadcast computation**: the verb simultaneously projects requirements for both subject and object positions.

These computational metaphors map directly onto the graph topologies we observe. Convergent computation produces narrow, deep trees with all edges pointing toward the root. Divergent computation produces wide trees with edges fanning outward from an early root. Broadcast computation produces flat structures with high branching at the root node.

6. Discussion

Our graph-topological framework offers several advantages over traditional typological approaches. First, by quantifying syntactic structure through continuous metrics rather than discrete categories, we can capture gradient differences both within and between languages. For example, Hindi and Japanese are both classified as SOV/head-final, but our topological analysis reveals that they differ substantially in projectivity and crossing number, reflecting Hindi's greater word-order freedom.

Second, the use of topological invariants (Betti numbers, Euler characteristic) provides a principled connection between the surface-level properties of word order and the deeper structural properties of syntactic

organization. The fact that β_1 correlates with morphological richness suggests a fundamental trade-off: languages can encode grammatical relations either through rigid word order (low β_1, projective topology) or through morphological marking (higher β_1, non-projective topology).

Third, the computational metaphors derived from topology (convergent, divergent, broadcast) have direct implications for NLP systems. Current transformer-based language models process all languages with the same architecture, but our analysis suggests that language-specific architectural choices might better match the natural topology of each language.

Limitations of this study include the use of simplified dependency representations and the focus on five languages. Future work should extend the analysis to a broader sample of languages, incorporate more sophisticated topological tools (persistent homology, sheaf theory), and validate the proposed metrics against psycholinguistic data on processing difficulty.

7. Conclusion

We have presented a framework for analyzing natural language syntactic structure through the lens of graph topology. By computing topological metrics and invariants over dependency graphs, we show that typologically different languages occupy distinct regions in topological space. The head directionality index, crossing number, and Betti numbers together provide a rich characterization of each language's "computational shape." Our analysis reveals a fundamental correspondence between topological structure and information processing strategy, suggesting that the topology of syntax is not arbitrary but reflects deep computational constraints on human language.

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